

BSc(Surveying) | RECO40II Dissertation

Adoption of Agentic AI in Quantity Surveying: Exploring Opportunities for Digital Transformation

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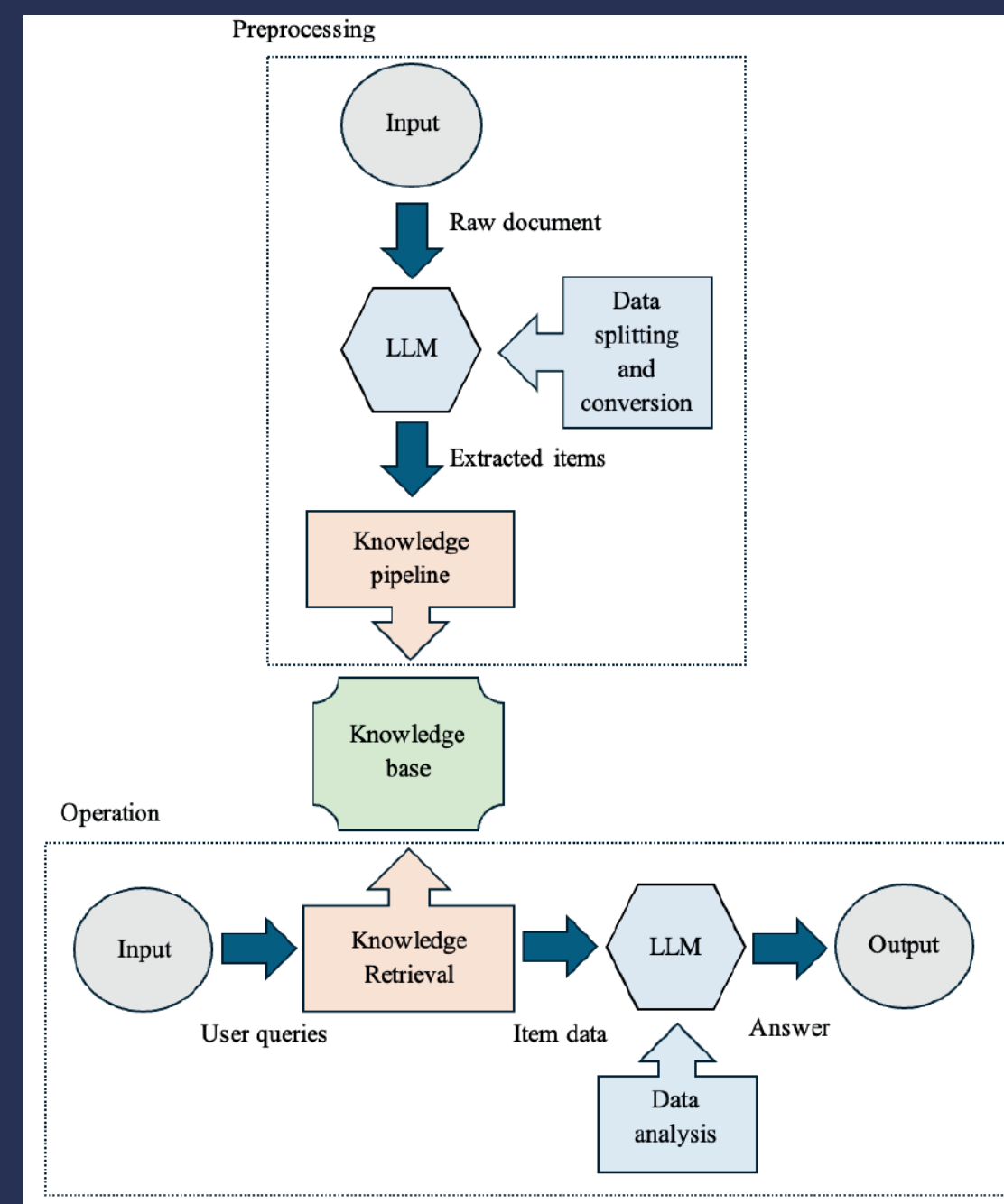
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Major Findings

The study found that agentic framework is capable of performing document-intensive tasks to produce consistent and reliable outputs. Three ready-to-use artefacts for the automation of routine quantity surveying tasks and modular frameworks were designed and developed using a Design Science Research (DSR) approach. The comparative advantage of agentic AI's capability over humans depends on the nature of task. Its capabilities are best demonstrated in the workflows that require database retrieval, information extraction, or manual searches. More especially, the Retrieval-Augmented Generation (RAG) system surpasses human performance in detecting subtle discrepancies and variations in documents.

From a perspective of cost-effectiveness, agentic AI presents a more economical solution. However, Qs are generally able to deliver more accurate results and completing most routine tasks faster than agentic frameworks. This performance gap is further amplified in tasks involving higher levels of complexity and professional judgement.

However, the current level of intelligence of Agentic AI remains insufficient to fully automate QS professional tasks. Thus, regular human review is required to ensure accuracy and reliability of outputs. Agentic AI should be positioned as a decision support and productivity enhancement tool, facilitating the automation of preliminary stages of routine tasks while maintaining professional control over final decisions rather than AI replacing Qs entirely.



Abstract

Construction 4.0 has brought a paradigm shift to the quantity surveying (QS) industry. Artificial intelligence (AI) has already been applied across a wide range of QS functions, including quantity takeoff (QTO), cost control, and risk management. However, the QS profession often involves multi-step, rule-based and evidence-driven workflows that require access to real-time data and contextual decision-making. Traditional AI frameworks, including single-prompt generative AI, generally exhibit limited functionality, reliability, and traceability in these dynamic project environments.

The study investigates these limitations by assessing the potential application of agentic AI, an emerging AI framework, using a Design Science Research (DSR) approach. Three Agentic AI workflow artefacts are designed and developed to automate routine tasks commonly undertaken by quantity surveyors. To evaluate their performance, the artefacts are tested using sample data acquired from QS professionals, and the results are compared with the performance of human practitioners in a series of workshop-based evaluation sessions.

The findings demonstrate the potential of agentic AI to automate repetitive, document-intensive QS tasks efficiently. However, regular human review and validation remain necessary to ensure reliability acceptability of outputs.

It is therefore recommended that agentic AI should serve as an intelligent assistant tool for quantity surveyors rather than serving as a complete replacement.

Performance of LLM models in three-rate bill test

